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Faculty of Computing

UNIVERSITI TEKNOLOGI MALAYSIA

UNIVERSITI TEKNOLOGI MALAYSIA
MID-TERM EXAMINATION SEMESTER II, SESSION 2024/2025

SUBJECT CODE : **SECR2242**

SUBJECT NAME : **COMPUTER NETWORKS**

SECTION : **1, 2, 3, 4**

TIME : **1700-1900**

DATE/DAY : **17 JUNE 2025 / TUESDAY**

VENUES : **N28, BK 1-4**

DO NOT OPEN THIS BOOKLET UNTIL YOU ARE TOLD TO DO SO

INSTRUCTIONS:

ANSWER ALL QUESTIONS.

(Please Write Your Name and Section in Your Answer Booklet)

Name	
IC/ISID No.	
Year / Course	
Section	
Lecturer Name	

WARNING: Disciplinary action will be taken against students who are found cheating during the examination.

This question paper consists of 6 (SIX) printed pages including this page.

[See next page]

PART A (16%) CLO1, PLO2

1. Basic Device Configuration

   (a) Briefly explain how subnetting and basic configuration help in organizing and securing an enterprise network. [3 marks]

   (b) Identify one potential risk if IP addressing is not planned properly. [2 marks]

2. Routing Concept and Static Route

   (a) What is the purpose of the ”next hop” in a static route configuration? [2 marks]

   (b) Explain the concept of ”administrative distance” and how it impacts routing decisions in a router with both static and dynamic routes [4 marks]

3. Dynamic Routing

   (a) You are a network administrator configuring RIPv2 on a Cisco router. You need to enable RIPv2, advertise the networks 192.168.1.0/24 and 10.0.0.0/16 and ensure that RIPv2 supports subnetting. [5 marks]

PART B (40%) CLO2, PLO2

1. Addressing and Subnetting IP Networks

   A medium-sized organization has been allocated the IP address block 192.168.10.0/24. The network must support four departments with the following number of required hosts:

   - HR: 50 hosts
   - IT: 30 hosts
   - Sales: 20 hosts
   - Management: 10 hosts

   The organization plans to implement static routing initially and may consider dynamic routing protocols later.

(a) Design an appropriate subnetting scheme to accommodate all four departments with minimum IP wastage. For each subnet, clearly specify:

- Subnet address
- Subnet mask (in CIDR and dotted decimal format)
- First usable host address
- Last usable host address
- Broadcast address

[7 marks]

(b) Explain why subnetting is important in organizational networks and how it supports efficient IP address allocation and routing decisions. [3 marks]

(c) Briefly compare static routing and dynamic routing protocols in the context of the given organization. Suggest one suitable dynamic routing protocol and justify your choice.

[2 marks]

2. Routing Concept and Static Route

(a) What is the role of EIGRP's metric in determining the best path? [2 marks]

(b) How would you configure a static route to send traffic to a destination network with the next-hop IP address 192.168.1.1? [2 marks]

(c) What is the role of a default route in a network, and how would you configure it on a router? [5 marks]

(d) How would you configure a static route on a router to ensure all traffic destined for the 192.168.100.0/24 network is forwarded to 192.168.1.2? [2 $\frac{1}{2}$ marks]

(e) How would you configure a static route to send all traffic to the 192.168.10.0/24 network through the interface GigabitEthernet0/2? [2 $\frac{1}{2}$ marks]

3. Dynamic Routing

(a) If RIPv2 is not working as expected, please rewrite the commands of troubleshooting steps according to the following issues

i. RIP may be running in version 1 (RIPv1), which does not support subnetting or authentication. [2 marks]

ii. By default, RIP performs automatic summarization at major network boundaries, which may cause incorrect routing. [2 marks]

iii. If networks are not advertised properly, they won't appear in the RIP routing table. [2 marks]

iv. RIP Neighbors Not Exchanging Routes [3 marks]

v. RIP updates take 30 seconds to exchange, causing slow convergence. [1 mark]

vi. Routing loops may occur due to outdated or incorrect routes. [3 marks]

PART C (44%) CLO1, PLO3

1. Basic Device Configuration

(a) **Scenario 1:**

You are a network technician tasked with configuring the basic settings for a small company's internal network. The company has three departments:

- Admin Department – 30 devices
- Marketing Department – 20 devices
- IT Department – 10 devices

The organization has been allocated the IP network 192.168.100.0/24.

Write the Cisco IOS commands to configure the router interfaces with IP addresses and enable them

[5 marks]

(b) Justify how your IP addressing and configuration choices demonstrate the application of scientific and technical skills in solving a real-world networking problem in terms of performance, security, and scalability.

[6 marks]

(c) Explain the primary purpose of configuring an IP address on a switch's VLAN 1 interface? [2 marks]

(d) Which action demonstrates technical problem-solving during basic device configuration in a secure network and justify it.

[2 marks]

2. Routing Concept and Static Route

(a) You have the following network topology:

i. RouterA (HR Dept): 192.168.1.1/24

ii. RouterB (IT Dept): 192.168.2.1/24

They are connected via a link 10.0.0.0/30, with RouterA on 10.0.0.1, and RouterB on 10.0.0.2.

Write the static routes needed on both routers so that each department can reach the other.

[8 marks]

(b) Your organization's router is connected to two paths toward a remote branch office:

- A primary connection that learns routes to the branch office using the EIGRP dynamic routing protocol
- A secondary backup connection that should only be used if the primary EIGRP route fails, with next-hop IP address 10.0.0.2

You are tasked with configuring a floating static route to provide failover to the backup link.

i. What is a floating static route, and how does it work in a network that uses EIGRP?

[2 marks]

ii. Write the command to configure a floating static route to the network 192.168.50.0/24 using 10.0.0.2 as the next-hop. [4 marks]

iii. Explain how the router determines when to switch from the EIGRP route to the floating static route. [2 marks]

3. Dynamic Routing

(a) You are a network administrator configuring dynamic routing using RIPv2 in a network with three routers (R1, R2, and R3). The network topology is as follows:

Network Topology :

- R1 - R2 Link: 192.168.1.0/30
- R2 - R3 Link: 192.168.2.0/30
- R1 LAN: 10.10.10.0/24
- R3 LAN: 172.16.1.0/24

Task:

i. Assign appropriate IP addresses to the interfaces of R1, R2, and R3 based on the given subnets. [3 marks]

ii. Configure RIPv2 on all three routers to enable dynamic routing. [3 marks]

iii. Rewrite the command to ensure that RIPv2 does not perform automatic summarization and supports subnetting for every routers. [1 mark]

iv. Rewrite the commands and elaborate it to verify the following setup

α) checking routing protocols [2 marks]

β) check the routing table [2 marks]

γ) check connectivity between R1 to R3 LAN and R3 to R1 LAN [3 marks]

Total is 100 marks